Mohammad Khamaysi

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Mixed-Signal and ASIC Design Engineer specializing in RTL-to-GDSII implementation of high-speed DSP systems for RF communication. Experienced in digital and mixed-signal VLSI flows, timing closure, and physical design. Proven ability to own full chip development from algorithm modeling to post-layout verification while optimizing performance, power, and signal integrity.

Education

M.Sc. Electrical and Electronic Engineering — Tel Aviv University (2024–Present) | GPA: 93
Excellence Scholarship Recipient, 2026

Thesis: RF Mixed-Signal Interpolator ASIC Design

Led the development of a multi-method RF interpolator ASIC from MATLAB modeling through RTL design and physical implementation. Designed a fixed-point Verilog architecture and executed full ASIC flow using Design Vision and Cadence Innovus, achieving a compact, low-power, fabrication-ready design.

- Achieved **~80 mRMS EVM** for QAM signals, validated using **Keysight VSA**
- Implemented full ASIC flow: **RTL, synthesis, place & route ,signoff**
- Optimized design to **~5 mW power consumption** using clock gating and architectural improvements
- Achieved compact layout with **~320 µm chip size**
- Designed and evaluated interpolation algorithms (**Linear, Lagrange, Cubic, Spline**) in MATLAB
- Closed timing and resolved setup/hold violations under physical design constraints

B.Sc. Electrical and Electronic Engineering — Tel Aviv University (2020–2024) | GPA: 84

Relevant Coursework: Introduction to VLSI (94), Digital Electronic Circuits (86), C Programming (89), Digital Logic (82), Electronic Devices (86), Data Structures (84), Signals and Systems (89), Random Signals and Noise (95), Integrated Analog IC Design (94), Digital VLSI Design (89), Mixed-Signal Circuits (99), VLSI Verification & Synthesis (100)

- VLSI Project: Designed a 4-bit ALU including logic, timing, area, and power considerations.
- Digital Circuits Project: Designed D-latch and flip-flop circuits optimized for timing.
- Analog IC Projects: Designed fully differential amplifier with CMFB.
- Mixed-Signal Project: Designed Flash ADC in Virtuoso with INL/DNL and power analysis.

Teaching Experience

- Tutor – Digital Electronic Circuits, Physics II, Introduction to VLSI
- Teaching Assistant – Energy Conversion Laboratory, Tel Aviv University
- Supervised 4 undergraduate students in final ASIC design projects

Technical Skills

- Programming & HDL: C, Python, MATLAB, Verilog, TCL
- EDA Tools: Synopsys Design Vision, Euclide, Verdi; Cadence Innovus, Virtuoso (Spectre); Mentor Calibre
- Concepts: ASIC Flow, Physical Design, Timing Closure, Power Optimization, Fixed-Point DSP, QAM Systems

Personal

- Oud player since 2018
- Designing electronic musical instruments
- Member of Oriental music band – Tel Aviv University
- Swimming

Languages: Arabic (Native), Hebrew (Fluent), English (Professional)